Yili Wen

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Research Interests

I design and build **wearable and robotic systems that restore and augment human movement**, working across **actuation and control, embedded sensing, and digital fabrication**. I prototype real hardware and evaluate it with the people it is built for. I want to extend this toward **assistive and rehabilitation robotics and exoskeletons**, and toward closing the loop between neural intent and adaptive physical assistance.

Education

2025 – Present	**Graduate Student, Computer Science — The University of Texas at Dallas** Design & Engineering for Making (DE4M) Lab, advised by Liang He. M.S. in Computer Science (Interactive Computing) expected **Spring 2027**.
2021 – 2025	**B.A. in Computation and Design — Duke Kunshan University** Dual-degree program with **B.A. in Interdisciplinary Studies, Duke University**. Undergraduate research advisor: Xin Tong.

Prior research positions: Research Assistant with Xin Tong (HCI Lab, Duke Kunshan University & Information Hub, HKUST Guangzhou, 2023–2025) and Karen Cochrane (Embodied Computing Lab, University of Waterloo, 2023–2025).

Publications

2026 [E.1]	***Yili Wen**, Yafei Ge, Xin Tong, Liang He. "PufFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication." (Work-in-Progress; presented). *ACM TEI 2026.*
2026 [W.2]	***Yili Wen**, Ceci Verbaarschot, Liang He. "From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients." (Workshop paper & poster; presented). *Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026, Barcelona.*
2026 [P.1]	Georgia Loewen, Karen Anne Cochrane, **Yili Wen**, Audrey Girouard. "Personas at Play: Understanding Disabled Player Experience to Expand Accessible Controller Design." (Poster). *Graphics Interface (GI) 2026.*
2025 [C.1]	***Yili Wen**, Jiaxin Wang, Hongni Ye, Liang He, Min Fan, Xin Tong. " 'I Want to Use Technology to Help Farmers in the Future': Enhancing Children's Understanding of Farming through Tangible User Interfaces and Embodied Interaction." (Presented). *ACM Interaction Design and Children (IDC) 2025.*
2024 [C.2]	Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. " 'I Keep Sweet Cats in Real Life, But What I Need in the Virtual World Is a Neurotic Dragon': Virtual Pet Design with Personality Patterns." *Chinese CHI 2024.*
2023 [W.1]	Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. "PetGen: Design and Generation of Virtual Pets." (Workshop). *IEEE ICME Workshops (ICMEW) 2023.*

Manuscripts In Preparation

–	***Yili Wen**, Ceci Verbaarschot, Liang He. "A Soft Grasp-Assist Exoskeleton for Spinal Cord Injury: Design and Intent-Driven Control." In preparation; targeting *IEEE Robotics and Automation Letters (RA-L).*
–	***Yili Wen**, Liang He. "PACER: Cable-Driven Guidance for Independent Running by Blind and Low-Vision Runners." In preparation; targeting *ACM IMWUT.*

Selected Research Projects

2025 – Present

Soft Grasp-Assist Exoskeleton for Spinal Cord Injury **("From Intention to Action")**

A wearable soft exoskeleton that restores grasping to people with spinal cord injury.

- Cast **three-chamber silicone finger actuators** from resin-printed molds, with a layer-jamming wrist brace for compliant grasp assistance during everyday tasks.
- Built the **ESP32** pneumatic control stack that drives the three chambers independently, and integrated simulated neural-intent input from a collaborating invasive-BCI group.
- Grounding the design in real rehabilitation needs through a clinician co-design study under an approved UT Dallas IRB protocol.

2025 – Present **PACER: Cable-Driven Wearable for Independent Blind & Low-Vision Running**
Blind and low-vision runners today depend on a sighted volunteer guide holding a tether, and what limits their running is guide availability rather than fitness. PACER is a body-worn cable-driven system that delivers the same directional pull the runner already trusts, so the run no longer requires a second person.

- Built the cable actuation modules and **ESP32** control stack that render direction as a pull, adapting the guide-rope tension cues runners already read, plus a custom object detector deployed on-device via Core ML in a SwiftUI companion app.
- Ran two formative studies that set the cue vocabulary, **five** interviews with a blind running community in Chongqing, China, and a two-population survey (**5** guide runners, **18** sighted runners) across a 19-situation taxonomy, which found the groups converge on *what* to do but diverge sharply on *how* it is conveyed.

2025 **PufFab: Shape-Transformable Rice Paper for Playful Food Fabrication** **(TEI '26)**
Modified a 3D printer and built the design tool for an edible-fabrication pipeline, where moistening and cutting control how each region of a flat rice-paper pattern puffs into a raised 2.5D form when fried.

2024 – 2025 **Adaptive Game-Controller Hardware for Upper-Limb Motor Disabilities** **(GI '26)**
Designed and fabricated the modular, remappable DIY controller prototypes that served as the physical platform for a study of disabled player experience.

2023 – 2024 **WooGu: Tangible System for Children's Agricultural Literacy** **(IDC '25)**
Built a tangible farm-to-table learning game pairing sensor-embedded 3D-printed farming tools with an augmented display; evaluated with **15 children** aged 5 to 12, improving their farming knowledge, engagement, and attitudes toward agriculture.

Teaching & Outreach

Summer 2026 **Instructor — TAST STEM Bridge & CS Outreach Summer Camps**
Department of Computer Science, UT Dallas. Six-week program (June–July 2026); designed and delivered hands-on lectures and build sessions.

2025 – 2026 **Teaching Assistant — Department of Computer Science, UT Dallas**
CS 4376 Object-Oriented Design (Spring 2026); CS 6326 Introduction to Human–Computer Interaction (Fall 2025).

Technical Skills

- **Fabrication & Prototyping:** soft-robotics fabrication (silicone molding/casting), SLA (resin) & FDM 3D printing, OpenSCAD / CAD, Blender, Figma.
- **Electronics & Embedded:** ESP32 / Arduino, soft pneumatic and DC-motor actuation, wireless (ESP-NOW).
- **Programming, ML & Vision:** Python, C/C++ (embedded), Swift/SwiftUI, JavaScript/p5.js, Java/Processing, R; custom object detection (YOLO / Ultralytics), on-device deployment (Core ML), OpenCV.
- **Research Methods & Analysis:** human-subjects studies (interviews, surveys, in-field evaluation), mixed methods, sensor-data analysis, statistics in R.

Awards

Center for the Study of Contemporary China (CSCC) Fellowship, Duke Kunshan University, 2023.
Dean's List, Duke Kunshan University — Fall 2022, Fall 2023, Spring 2024, Fall 2024.